

Long line 2.0 — Solution

Y25 (2025) — English translation

A1

0.40 pts

Express the voltage across resistor $U_R(t)$, and the current through it $I_R(t)$ in terms of $U_l(x, t)$, $U_r(x, t)$, $I_l(x, t)$, $I_r(x, t)$ and $\mathcal{E}(t)$. Obtain an expression connecting $U_l(0, t)$ and $U_r(0, t)$. The expression may contain R , Z , $\mathcal{E}(t)$.

The voltage drop across the resistor is $U_R(t) = \mathcal{E}(t) - (U_l(0, t) + U_r(0, t))$ and the current is $I_R(t) = \frac{U_R(t)}{R} = I_l(0, t) + I_r(0, t) = \frac{1}{Z} (U_r(0, t) - U_l(0, t))$. From here we get:

$$\frac{\mathcal{E}(t)}{R} - \frac{U_l(0, t)}{R} - \frac{U_r(0, t)}{R} = \frac{U_r(0, t)}{Z} - \frac{U_l(0, t)}{Z} \implies$$

Any equivalent response forms are valid, but the one written above is most convenient for analysis.

Answer:

$$\frac{\mathcal{E}(t)}{2} - U_r(0, t) = \frac{R - Z}{R + Z} \left(\frac{\mathcal{E}(t)}{2} - U_l(0, t) \right)$$

A2

1.10 pts

Find the time dependence of voltage $U_{\text{OSC}}(t)$ on an oscilloscope.

Let us denote it by $U_n = U_0 \left(1 - \left(\frac{R - Z}{R + Z} \right)^n \right)$ for convenience. At the initial time $U_l(0, 0) = 0$, $U_r(0, 0) = U_1/2$. A wave propagating to the right reaches the oscilloscope in time L/v and is reflected from the edge of the two-wire line. After another time L/v , the wave traveling to the left returns to the resistor, and then $U_l(0, 2L/v) = U_1/2 \implies U_l(0, 2L/v) = U_2/2$. Next, the process of re-reflection of the traveling waves is repeated, and after time $2nL/v$ the voltage $U_l(x = 0)$ reaches $U_n/2$, and the voltage $U_r(x = 0) = U_{n+1}/2$. From here we come to the final result:

Answer:

$$U_{\text{OSC}}(t) = U_0 \left[1 - \left(\frac{R - Z}{R + Z} \right)^{\lfloor \frac{vt}{2L} + \frac{1}{2} \rfloor} \right]$$

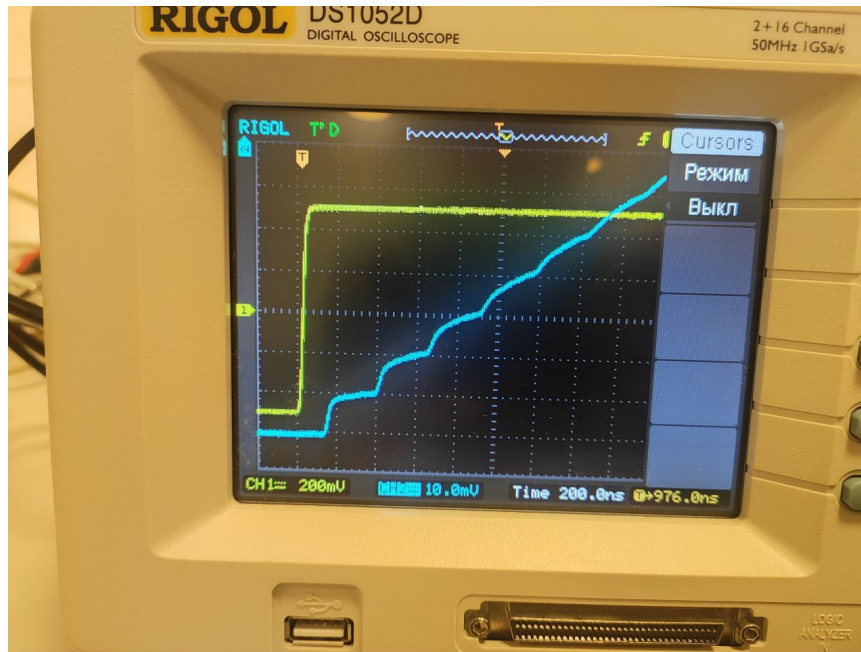
A3

0.70 pts

By applying a rectangular signal from the generator, remove the dependence of voltage on time. Record the times t_n when the n voltage step occurred, and the values of voltage U_n immediately after.

The voltage step ladder appears on time scales of order $L/c \sim 10^{-8} \dots 10^{-7}$ s, so it is visible only on time scales of order ~ 100 ns. Although the ladder becomes clearer with more cable sections, the number of unlubricated “steps” is still small, so measurements must be taken with a large number of cable sections. A typical view of the ladder is shown in the photo below:

A typical view of the ladder is shown in the photo below:



A4

2.80 pts

Repeat the previous step, connecting more cable sections (from 2 to 5 inclusive).

$m = 1$ cable section:

$n \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ (2n - 1) \ m \ 1 \ 3 \ 5 \ 7 \ 9 \ 11 \ 13 \ 15 \ t_n, \text{ ns}$ 44 95.2 151 205 258 314 365 424 $\Delta t_n, \text{ ns}$ 18 17
19 18 18 19 17 20 $U, \text{ mV}$ -479 -470 -460 -451 -442 -433 -424 -415 $\Delta U, \text{ mV}$ 3 3 3 3 3 3 3 u -0.479 -0.47
-0.46 -0.451 -0.442 -0.433 -0.424 -0.415 Δu 0.003 0.003 0.003 0.003 0.003 0.003 0.003 0.003

$m = 2$ cable section:

$n \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ (2n - 1) \ m \ 2 \ 6 \ 10 \ 14 \ 18 \ 22 \ 26 \ 30 \ t_n, \text{ ns}$ 28 130 238 338 448 548 656 756 $\Delta t_n, \text{ ns}$ 35
34 36 33 37 33 36 33 $U, \text{ mV}$ -479 -469 -459 -450 -440 -431 -422 -413 $\Delta U, \text{ mV}$ 3 3 3 3 3 3 3 u -0.479
-0.469 -0.459 -0.45 -0.44 -0.431 -0.422 -0.413 Δu 0.003 0.003 0.003 0.003 0.003 0.003 0.003 0.003

$m = 3$ cable section:

$n \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ (2n - 1) \ m \ 3 \ 9 \ 15 \ 21 \ 27 \ 33 \ 39 \ 45 \ t_n, \text{ ns}$ 50 210 364 520 680 834 988 1150 $\Delta t_n, \text{ ns}$ 52
53 51 52 53 51 51 54 $U, \text{ mV}$ -489 -478 -469 -459 -449 -440 -430 -421 $\Delta U, \text{ mV}$ 4 4 3 3 3 3 3 u -0.489
-0.478 -0.469 -0.459 -0.449 -0.44 -0.43 -0.421 Δu 0.004 0.004 0.003 0.003 0.003 0.003 0.003 0.003

$m = 4$ cable section:

$n \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ (2n - 1) \ m \ 4 \ 12 \ 20 \ 28 \ 36 \ 44 \ 52 \ 60 \ t_n, \text{ ns}$ 80 284 496 704 906 1120 1320 1530 $\Delta t_n, \text{ ns}$
69 68 71 69 67 71 67 70 $U, \text{ mV}$ -489 -479 -468 -458 -449 -439 -430 -420 $\Delta U, \text{ mV}$ 3 3 4 3 3 3 3 u -0.489
-0.479 -0.468 -0.458 -0.449 -0.439 -0.43 -0.42 Δu 0.003 0.003 0.004 0.003 0.003 0.003 0.003 0.003

$m = 5$ cable sections:

$n \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ (2n - 1) \ m \ 5 \ 15 \ 25 \ 35 \ 45 \ 55 \ 65 \ 75 \ t_n, \text{ ns}$ 104 360 616 876 1120 1390 1660 1900
 $\Delta t_n, \text{ ns}$ 84 85 85 87 81 90 90 80 $U, \text{ mV}$ -489 -479 -469 -459 -449 -439 -429 -420 $\Delta U, \text{ mV}$ 3 3 3 3 3 3 3
 u -0.489 -0.479 -0.469 -0.459 -0.449 -0.439 -0.429 -0.42 Δu 0.003 0.003 0.003 0.003 0.003 0.003 0.003 0.003

A5

1.00 pts

Neglecting the length of the cables used to connect the generator, linearize the dependences $u(n, m)$ and $t(n, m)$. Plot the linearized dependence $u(n, m)$ and determine the resistance R of the supplied resistor if the characteristic impedance of the cable is $50 \ \Omega$. Plot a linearized graph of $t(n, m)$ and determine the speed v of wave propagation in the coaxial cable. Give the errors of the results.

For m connected cable sections we will have:

$$t(n, m) = (2n - 1)mL_0/v, \quad L_0 = 5 \text{ m.}$$

Linearization – $t((2n - 1)m)$ with slope L_0/v . Since the voltage on the unlubricated part of the ladder is

much less than the voltage U_0 , we can write for it as a first approximation:

$$U_n = U_0 \left(1 - \left(\frac{R - Z}{R + Z} \right)^n \right) \approx \frac{2nZ}{R} U_0.$$

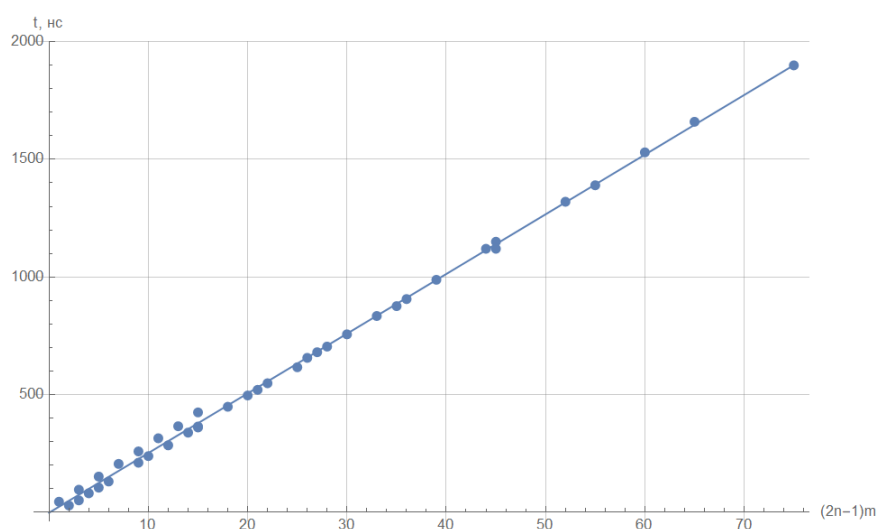
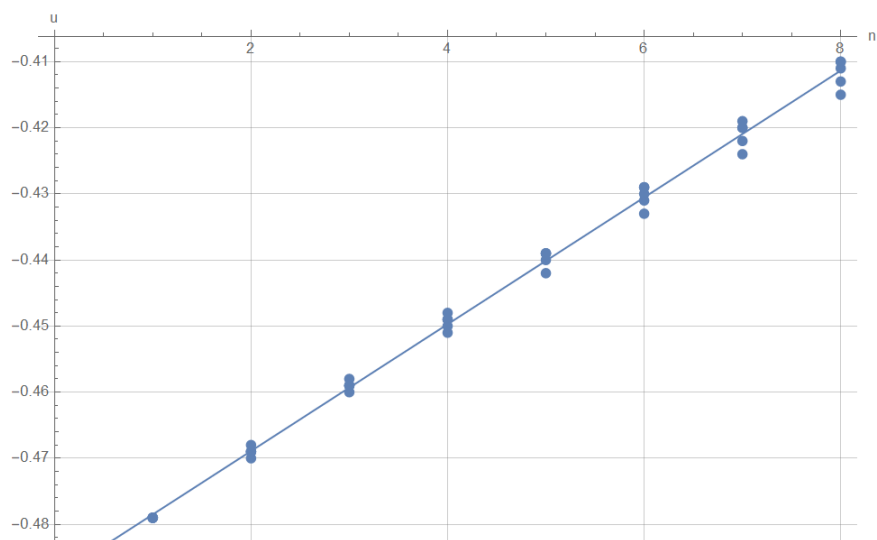
Therefore, a convenient linearization is $u(n)$, and its slope is equal to $2Z/R$.

The angular coefficient of dependence $u(n)$ is equal to $\frac{2Z}{R} = 9.6 \cdot 10^{-3}$, we estimate the error as $\varepsilon_R \sim 2\Delta u / (u_{\max} - u_{\min}) \approx 9\%$, then:

The angular coefficient of dependence $t(m(2n - 1))$ is equal to $\frac{L_0}{v} = 25.4$ ns, we estimate the error as $\varepsilon_v \sim 2\Delta t / (t_{\max} - t_{\min}) \approx 9\%$, then:

Answer:

$$R = (10.4 \pm 0.9) \text{ k}\Omega$$



B1

1.50 pts

Neglecting the intrinsic capacitance of the cables used to connect the generator and oscilloscope, propose a circuit that allows you to determine the capacitance of a coaxial cable per unit length. Take measurements for different numbers of cable sections and determine its capacity per unit length. Knowing the charac-

teristic impedance of the cable, find its inductance per unit length. Estimate the error of the obtained results.

The most convenient connection option:

We will measure the ratio of the signal amplitudes on the cable (capacitor) and the entire circuit. Since in such a chain:

$$\left| \frac{U_{RC}}{U_C} \right|^2 = \left| \frac{R + \frac{1}{i\omega C}}{\frac{1}{i\omega C}} \right|^2 = 1 + 4\pi^2 R^2 L_0^2 C^2 \cdot m^2 f^2.$$

We use the linearization $\left| \frac{U_{RC}}{U_C} \right|^2 (m^2 f^2)$. The input voltage is equal to $U_{RC} = 1$ V throughout. Since the error in the resistor resistance value already reaches 5 – 10 %, we can neglect the oscilloscope error. $m = 1$ cable section:

$m = 1$ cable section:

f , kHz 10 20 30 40 50 60 70 U_C , mV 938 810 695 585 504 436 378 $m^2 f^2$, 10^9 Hz^2 0.1 0.4 0.9 1.6 2.5 3.6 4.9 $(1 \text{ V}/U_C)^2$ 1.14 1.52 2.07 2.92 3.94 5.26 7.00

$m = 2$ cable section:

f , kHz 10 20 30 40 45 50 55 60 U_C , mV 821 590 446 344 315 282 260 234 $m^2 f^2$, 10^9 Hz^2 0.4 1.6 3.6 6.4 8.1 10.0 12.1 14.4 $(1 \text{ V}/U_C)^2$ 1.48 2.87 5.03 8.45 10.08 12.57 14.79 18.26

$m = 3$ cable section:

f , kHz 5 10 15 20 25 30 35 U_C , mV 889 723 552 448 368 314 274 $m^2 f^2$, 10^9 Hz^2 0.23 0.90 2.03 3.60 5.63 8.10 11.03 $(1 \text{ V}/U_C)^2$ 1.27 1.91 3.28 4.98 7.38 10.14 13.32

$m = 4$ cable section:

f , kHz 5 10 13 16 19 22 25 U_C , mV 823 600 505 427 368 323 286 $m^2 f^2$, 10^9 Hz^2 0.40 1.60 2.70 4.10 5.78 7.74 10.00 $(1 \text{ V}/U_C)^2$ 1.48 2.78 3.92 5.48 7.38 9.59 12.23

$m = 5$ cable sections:

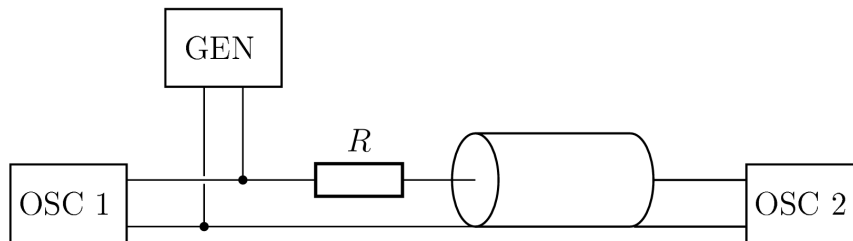
f , kHz 3 6 8 10 12 14 16 U_C , mV 885 711 596 523 448 388 347 $m^2 f^2$, 10^9 Hz^2 0.23 0.9 1.6 2.5 3.6 4.9 6.4 $(1 \text{ V}/U_C)^2$ 1.28 1.98 2.82 3.66 4.98 6.64 8.31

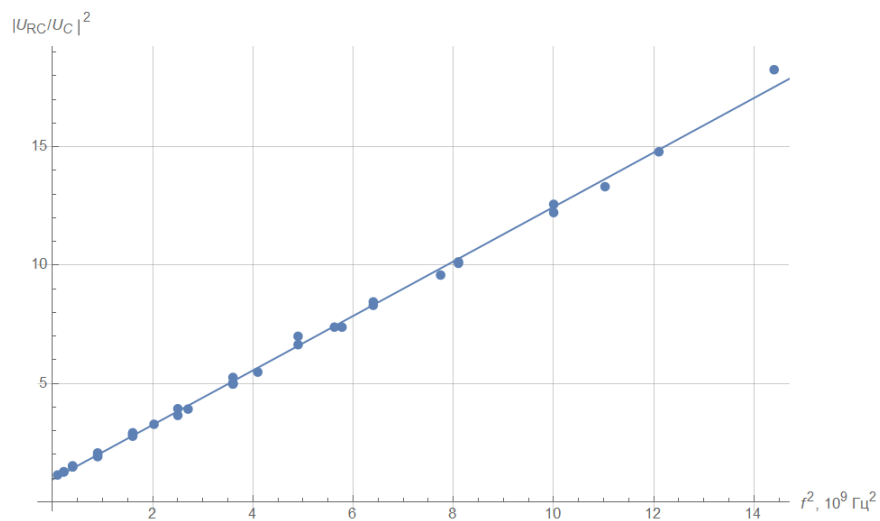
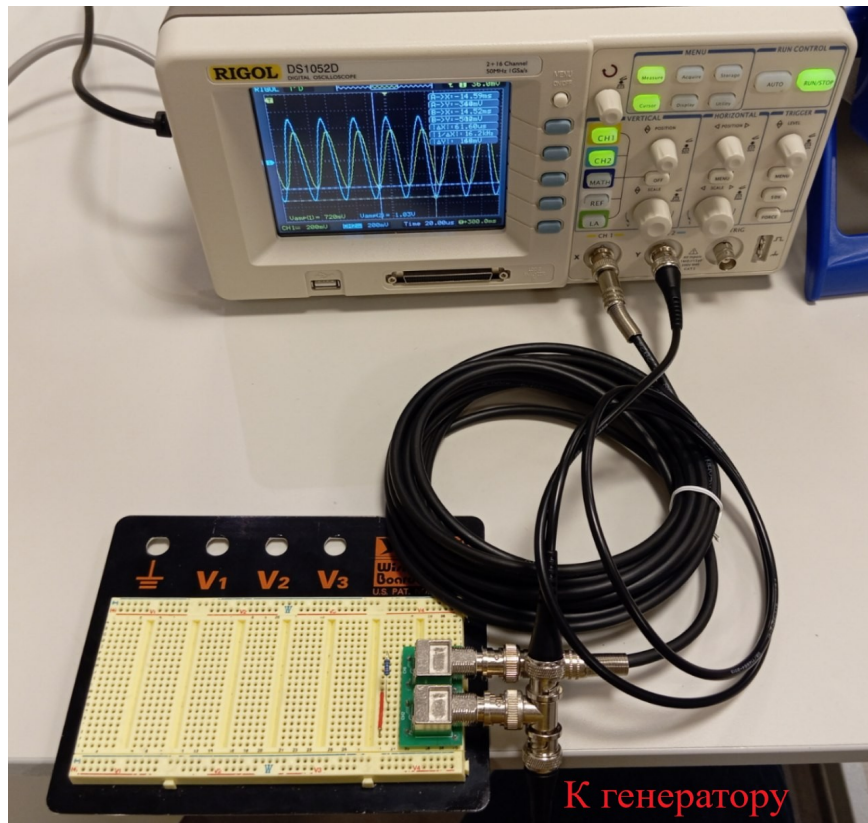
The slope is $4\pi^2 R^2 L_0^2 C^2 = 1.15 \cdot 10^{-9} \text{ Hz}^{-2}$, hence $C = 1.04 \cdot 10^{-10} \text{ F/m}$. We estimate the error as $\sim 10\%$, then:

Since $Z = \sqrt{\frac{\mathcal{L}}{C}}$, the cable inductance per unit length is $\mathcal{L} = Z^2 C = 2.6 \cdot 10^{-7} \text{ H/m}$.

Answer:

$$C = (1.04 \pm 0.10) \cdot 10^{-10} \text{ F/m}$$





C1

1.00 pts

Having supplied a rectangular signal from the generator, remove the dependence of voltage on time, namely, find the moments of time t_n when the n stepwise increase in voltage occurred, and the voltage values U_n immediately after that.

n	1	2	3	4	5	6	7	8	$2n - 1$	1	3	5	7	9	11	13	15	t_n , ns	24	134	232	338	436	548	648	750	Δt_n , ns	110	98	106			
U , mV	-488	-468	-447	-427	-408	-389	-371	-353	ΔU , mV	20	21	20	19	19	18	18	u	-0.488	-0.468	-0.447	-0.427	-0.408	-0.389	-0.371	-0.353	Δu	0.02	0.021	0.02	0.019	0.019	0.018	0.018

Find the speed v of signal propagation in the twisted pair and its characteristic impedance Z . Give the errors in the results obtained.

The angular coefficient of dependence $u(n)$ is equal to $\frac{2Z}{R} = 1.93 \cdot 10^{-2}$, we estimate the error as $\varepsilon_Z \sim 2\Delta u / (u_{\max} - u_{\min}) \approx 15\%$, then:

The angular coefficient of dependence $t(2n - 1)$ is equal to $\frac{L_0}{v} = 51.8$ ns, we estimate the error as $\varepsilon_v \sim 2\Delta t / (t_{\max} - t_{\min}) \approx 9\%$, then:

Answer:

$$R = (100 \pm 20) \, \Omega$$

